Michael Liu

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Education

University of Waterloo — *Sept 2025 - Apr 2030*
BCS in Computer Science, Artificial Intelligence Specialization

- Major Average: 93.1% | Cumulative Average: 92.7% | Student ID: 21177966
- Activities: Varsity Cross-Country, Varsity Track, WAT.ai

Job Experience

Machine Learning Research Assistant — *Toronto, ON*
The Hospital for Sick Children (SickKids) — *May 2026 - Aug 2026*

- Researching reaction-based GFlowNet models for molecular design and drug discovery.

Machine Learning Scientist — *Waterloo, ON*
WAT.ai — *Sept 2025 - Feb 2026*

- Collaborated with a small team of students on a medical imaging project focused on improving the segmentation of microaneurysms in fundus imaging.
- Co-authored a research paper detailing our methodology and results.

Tennis Instructor — *Unionville, ON*
Unionville Tennis Club — *Apr 2022 - Aug 2025*

- TPA Certified Instructor, 500+ Hours On-Court.

Projects

CourtVision: Computer Vision for Tennis Analytics — *github/tennis-cv-analysis*
OpenCV, PyTorch, YOLO, MediaPipe, XGBoost, Pandas, SciPy

- Built a computer vision system for tennis match analysis including player/ball tracking, court keypoint detection, and shot detection from broadcast camera footage.
- Implemented pose-based shot classification and computed player and ball speed statistics displayed through annotated match video output.

ArXtract: AI Search Engine for ML Research — *arxtract-cxc.vercel.app*
React, TypeScript, FastAPI

- Built a full-stack ML research tool using retrieval-augmented generation to query, rank, and summarize arXiv papers.
- Engineered semantic search using vector embeddings and cosine similarity to extract relevant sections from machine learning research papers based on user prompts.

HydraLA-Net Optimization for Microaneurysm Segmentation — *github/fundus-image-segmentation*
PyTorch, NumPy, Albumentations, Pandas, scikit-learn, Streamlit

- Built PyTorch training pipeline and trained HydraLA-Net (U-Net variation) segmentation models for retinal lesion detection.
- Conducted ablation studies on preprocessing and loss-function design to improve small lesion segmentation.

Technical Skills

Programming Languages:	Python, SQL, C, TypeScript
Data/ML:	PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, LLM/GenAI, OpenCV, YOLO
Databases/Tools:	PostgreSQL, MySQL; Git, GitHub; LaTeX; Excel, Tableau
Frontend:	React, HTML, CSS